

# Regime-Adaptive Multimodal Transformer for Cross-Market Equity Forecasting

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**Abstract**—We test whether a regime-adaptive multimodal transformer (RAMT) can rank NIFTY 200 constituents cross-sectionally on a 21-day horizon. RAMT fails: predictions collapse ( $\sigma \approx 0.0015$  across names), validation IC turns negative, and a 60-line LightGBM diagnostic on the same features produces  $IC = +0.021$ . We replace RAMT with two alternatives: (a) plain 21-day momentum gated by a 3-state Hidden Markov Model on the index, and (b) Chronos-T5-Small fine-tuned with LoRA ( $r=8$ ) on the same 10 features. Across the 2024–2026 out-of-sample window, Foundation-Only Chronos-LoRA delivers Sharpe 1.34, the strongest single component; Momentum+HMM delivers Sharpe 0.83 with a more conservative drawdown profile; the regime-gated Triple-Expert hybrid *underperforms* both at Sharpe 0.54. Across four historical HMM ablation windows (2008–2010, 2010–2012, 2013–2015, 2024–2026) the regime gate saves 9.4pp of drawdown in the 2008 crisis and turns a Sharpe of  $-3.0$  into  $+0.79$  in 2010–2012 but caps upside in clean bulls. Our three findings: (1) simpler architectures beat deeper ones at sub-million-parameter scales on this universe; (2) the RAMT failure is architectural, not a feature problem (LightGBM and momentum found signal on the same panel); (3) the HMM is *conditional insurance*, not always-on alpha.

**Index Terms**—quantitative finance, factor investing, hidden markov models, foundation models, LoRA, Chronos-T5, regime detection, NIFTY 200, ablation

## I. INTRODUCTION

The original hypothesis of this project was that a multimodal transformer with explicit regime cross-attention and a tournament ranking loss would extract cross-sectional alpha from NIFTY 200 names that simpler factor models miss. The hypothesis is rejected by the data. After two attempts at training the transformer (RAMT) we observed prediction-variance collapse and negative validation IC. A diagnostic with plain LightGBM on the same ten features produced statistically meaningful information coefficient, ruling out feature adequacy as the cause. The remainder of the project tests two simpler alternatives and a foundation-model fine-tune.

The contribution of this paper is empirical: we report a complete ablation across one ML-only baseline (Momentum+HMM), one transformer-from-scratch attempt (RAMT), one foundation-model fine-tune (Chronos-T5-Small with LoRA adapters), and two interaction modes between them, on the same 2024–2026 out-of-sample window with identical friction and risk-control rules. We then report a

separate four-window study isolating HMM regime sizing against flat sizing.

## II. RELATED WORK

Cross-sectional momentum on equities was established by Jegadeesh and Titman [?] and refined into the global “everywhere” formulation by Asness, Moskowitz, and Pedersen [?]. Hidden Markov Models for regime detection follow Hamilton’s regime-switching framework [?], extended to multivariate financial settings by Guidolin and Timmermann [?]. Transformer architectures originate from Vaswani et al. [?] and were specialized for time-series forecasting in Lim et al. [?]. Our Phase 3 substitutes a pre-trained foundation model: Chronos [?] treats time-series tokenization with the T5 architecture, and we adapt it parameter-efficiently using LoRA [?], which constrains updates to low-rank decompositions  $W = W_0 + BA$  with  $B \in \mathbb{R}^{d \times r}$ ,  $A \in \mathbb{R}^{r \times k}$ ,  $r \ll \min(d, k)$ .

## III. DATA AND FEATURES

**Universe.** A static 2026 NIFTY 200 snapshot. We acknowledge survivorship and constituent-drift bias as a limitation rather than a fix.

**Sources.** Daily OHLCV per ticker via `yfinance`; index series for `^NSEI` for the regime feed and benchmark; sector mapping is a hand-curated table in `features/sectors.py`.

**Features (10).** Per ticker per date, lagged at  $T-1$  to enforce information frontiers: 21-day return (`Ret_21d`), 5-day return, realized volatility (21-day), volume z-score, price relative to 50-day SMA, price relative to 200-day SMA, 14-day RSI, MACD histogram, NIFTY 21-day return (macro feed), NIFTY realized volatility (macro feed). The label is 21-day forward sector-neutral alpha (`Sector_Alpha`).

**Modalities (4).** Price, technicals, volume, macro. The HMM is fit on the NIFTY index alone (3-state Gaussian, fit by Baum–Welch).

**Manifest.** `data/manifest.csv` ships 201 parquet entries with md5 hashes; `scripts/check_pipeline_health.py` --verify-manifest re-hashes them on demand for reproducibility.

#### IV. PHASE 1: BASELINE MODELS

We first tested daily-return prediction with XGBoost and an LSTM. Both failed: directional accuracy 49.4% (XGBoost) and 48.9% (LSTM) on the walk-forward test set; mean information coefficient  $-0.061$  and  $-0.041$  respectively (results/backtesting/phase1\_baselines/). The label was the bug, not the algorithm: daily returns at this granularity are dominated by noise. We pivoted to 21-day forward sector-neutral alpha for Phases 2 and 3.

#### V. PHASE 2: RAMT ARCHITECTURE AND FAILURE

##### A. Architecture

RAMT comprises six modules: (1) a MultimodalEncoder for the four modalities; (2) standard sinusoidal positional encoding; (3) RegimeCrossAttention, where the regime embedding is the query; (4) a 3-expert MoE block (bull, high-vol, bear flavor) followed by transformer layers; (5) dual prediction heads for monthly and daily horizons; and (6) a tournament ranking loss with an auxiliary daily MSE anchor.

##### B. Training Setup

Walk-forward, 6-month refresh, sequence length 30. After speed and overfitting fixes (workers = 4, cache = 250, MoE collapsed to a single expert), the final architecture has 131,298 parameters across 2 transformer layers and 8 heads.

##### C. Failure Analysis

Across 25 epochs and four walk-forward folds, validation correlation to Sector\_Alpha went *negative* ( $-0.036$ ) and worsened monotonically. Per-date prediction standard deviation across the 200 names collapsed to  $\sigma \approx 0.0015$ . The mechanism: the tournament ranking loss has a fixed-point at zero scores — once predictions collapse, pairwise margins vanish, gradients vanish, and the model has no signal to escape. We later raised the daily-MSE anchor weight from 0.05 to 0.20, restored capacity to 131k parameters, and re-ran 7 epochs; the negative-correlation curve was already monotonic and we stopped.

Final RAMT metrics on the blind 2024–2026 window: DA = 47.2%, RMSE = 0.072, mean IC =  $-0.016$ , backtested Sharpe = 0.485, MaxDD =  $-6.4\%$  (results/models/ramt/ramt\_metrics.json).

##### D. Diagnostic that motivated the pivot

A 60-line LightGBM script on the same 10 features and the same train/test split produced IC =  $+0.021$ , IR =  $+6.41$ , top-5 positive-alpha rate 53.8%. Pure 21-day momentum without any model produced IC =  $+0.0065$ , top-5 spread  $+1.59\%$  per month, top-5 positive rate 56.3%. Both beat RAMT. The features carried signal; the transformer did not extract it.

#### VI. PHASE 3: FOUNDATION MODEL WITH LORA

##### A. Architecture

We freeze the pre-trained Chronos-T5-Small backbone ( $\sim 190\text{M}$  parameters) and attach LoRA adapters with rank  $r=8$  to the attention projection matrices, training only the adapters and a small ranking head — together approximately 0.1% of the backbone’s parameter count. Input is the same ( $B, 30, 10$ ) feature tensor used by RAMT. Output is a scalar alpha score per ticker per rebalance date.

##### B. Training Setup

LoRA  $r=8$ ,  $\alpha=16$ , dropout 0.1. Optimizer AdamW, learning rate  $1e-4$ , batch size 64, walk-forward training. Reference scripts: models/lora\_experiment/chronos\_lora\_v2.py, scripts/train\_chronos\_lora.py. The trained adapter is committed to results/models/lora/chronos\_lora\_adapter.pt.

##### C. Results

Out-of-sample directional accuracy DA = 47.5% (LoRA v1) / 47.0% (LoRA v2), but mean IC turns positive ( $+0.0090$  /  $+0.0022$ ) — a sign change relative to RAMT. More importantly, prediction standard deviation no longer collapses ( $\sigma_{\text{pred}} \approx 4.7e-3$  for LoRA versus  $\sim 1.5e-3$  for RAMT, a  $\sim 3\times$  widening that allows ranking to actually separate names). Backtested Sharpe on 2024–2026 reaches 1.34 (Foundation-Only), the strongest single-component result in the project. See Table I.

#### VII. FINAL STRATEGY: MOMENTUM + HMM + SECTOR CAP

The currently-deployed strategy is intentionally simpler than the ablation winner. Each rebalance date (every 21 trading days):

- 1) Sort the 200 names by 21-day return.
- 2) Take top 5 with at most one name per NSE sector.
- 3) Size the book by the current HMM regime: 100% / 50% / 20% for Bull / High-Vol / Bear.
- 4) Apply per-stock 7% stop and a 15% portfolio drawdown killswitch.
- 5) Charge friction at 0.22% of traded notional.

Out-of-sample (2024-01-10 to 2026-01-27, 2-year window): Sharpe = 0.83, CAGR = 13.5%, MaxDD =  $-18.7\%$ , win rate 64%. Versus NIFTY buy-and-hold over the same calendar (Sharpe 0.65, CAGR 7.7%, MaxDD  $-15.8\%$ ):  $+0.18$  Sharpe,  $+5.8\text{pp}$  CAGR,  $-2.9\text{pp}$  on max drawdown.

#### VIII. ABLATION

##### A. Cross-component ablation (2024–2026 window)

The two interpretable component-removal cells are aliases of existing scenarios because, by construction, removing HMM from the Triple-Expert collapses to the 50/50 hybrid, and removing Chronos collapses to Momentum+HMM. The third cell (Chronos+HMM, no

TABLE I  
NET-OF-FRICTION PERFORMANCE ON THE 2024–2026 OUT-OF-SAMPLE WINDOW.

| Variant                                  | Sharpe                   | CAGR  | MaxDD  |
|------------------------------------------|--------------------------|-------|--------|
| Momentum + HMM (production)              | 0.83                     | 13.5% | -18.7% |
| RAMT transformer                         | 0.49                     | n/a   | -6.4%  |
| Chronos-LoRA Foundation-Only             | <b>1.34</b>              | 23.5% | -16.0% |
| Simple Hybrid (50/50 Mom + Chronos)      | 0.91                     | 22.8% | -11.1% |
| Triple-Expert (Mom + Chronos + HMM)      | 0.54                     | 8.2%  | -13.8% |
| <i>Component-removal cells:</i>          |                          |       |        |
| Hybrid – HMM (alias of Simple Hybrid)    | 0.91                     | 22.8% | -11.1% |
| Hybrid – Chronos (alias of Mom+HMM prod) | 0.83                     | 13.5% | -18.7% |
| Hybrid – Momentum                        | not run — module missing |       |        |

momentum) is honestly recorded as *not run*: the in-repo `models/hybrid_backtester.py` module referenced from `scripts/run_diagnostic_ablation.py` is absent in the current working tree, and we chose not to fabricate a number for the table.

#### B. HMM regime-sizing ablation across four historical windows

TABLE II  
HMM-CONDITIONED VS FLAT SIZING, FOUR WINDOWS.

| Window           | HMM Sharpe | Flat Sharpe | HMM DD | Flat DD |
|------------------|------------|-------------|--------|---------|
| 2008–2010 (GFC)  | 0.25       | 0.17        | -43.2% | -52.7%  |
| 2010–2012        | 0.79       | -3.00       | -12.2% | -30.1%  |
| 2013–2015        | 0.84       | 0.59        | -33.2% | -33.2%  |
| 2024–2026 (bull) | 0.66       | 1.35        | -18.7% | -19.7%  |

The pattern is consistent across four windows: HMM gating helps in volatile and crisis windows (saves 9.4pp drawdown in 2008, rescues a  $-3.0$  Sharpe in 2010–2012) and *hurts* in clean bull markets (cuts Sharpe from 1.35 to 0.66 in 2024–2026). Treating HMM as conditional insurance rather than always-on alpha is the central architectural lesson of the project.

#### IX. LIMITATIONS

**Survivorship bias.** The universe is a 2026 NIFTY 200 snapshot, not the historically reconstructed index membership. Although the panel is time-aligned, the cross-section excludes names that were in the index but later delisted.

**Window length.** The headline 2024–2026 Phase 3 numbers are from a single two-year out-of-sample window. The strategy has not been tested through a 2008-style or 2020-style crash in this configuration; the four-window HMM study is the closest stress test.

**Stop-loss is a no-op in places.** The 7% per-stock stop and 15% portfolio killswitch were rarely triggered in the 2024–2026 window, so the backtest does not strongly exercise that path.

**Friction model.** Costs are modeled as a fixed 0.22% of traded notional per rebalance. Real Indian small/mid-cap impact is often higher and varies with name and size; this is a known optimistic bound.

**Static feature set.** We did not search over feature subsets or include text/sentiment features in the final ablation, although a FinBERT sentiment pathway exists in the repository.

**Incomplete component-removal ablation.** The Hybrid – Momentum ablation cell, which would isolate the contribution of Chronos-LoRA predictions without the momentum ranking signal, could not be computed at submission time due to a missing backtesting module dependency. This cell is recorded as *not run* in the ablation summary JSON and represents an open validation gap. All other component-removal cells are populated from real experimental artifacts.

#### X. CONCLUSION

Three findings:

*Simpler beat deeper.* At our data and parameter scale, plain 21-day momentum outperformed a 131k-parameter regime-aware transformer trained end-to-end. Foundation-Only Chronos-LoRA, which leans on a 190M-parameter pre-trained backbone with only  $\sim 0.1\%$  trainable, then beat both. The inflection happened at the foundation-model boundary, not by adding architectural complexity to the from-scratch model.

*The RAMT failure was architectural.* The same 10 features driven into LightGBM produced positive IC; driven into 21-day momentum produced positive top-5 spread. The transformer’s tournament ranking loss collapsed to a fixed-point and the gradients went to zero. We document this honestly rather than masking it with longer training.

*HMM is conditional insurance.* Across four historical windows the HMM regime gate adds Sharpe and reduces drawdown in crisis and high-volatility windows, and subtracts Sharpe in clean bulls. The honest deployment policy is to gate *only* during volatile regimes; using HMM as an always-on multiplier is what produced the Triple-Expert Sharpe of 0.54. The natural next step is a learned regime-conditional gate that defers to Foundation-Only in calm regimes and to HMM-protected sizing in volatile ones, with the gate trained on regime-conditional out-of-sample Sharpe rather than hand-tuned weights.

#### REFERENCES